



--

--	--	--	--	--

--	--	--	--

9648 / 02

2 hours

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

This document consists of 2 sections:

Answer **ALL** questions.
Answers are to be written in the spaces provided.

Answer **ONE** question.
Write your answers on the separate writing papers
provided.

For Examiner's Use	
Structured Questions	
1	/ 9
2	/ 10
3	/ 11
4	/ 10
5	/ 10
6	/ 10
7	/ 10
8	/ 10
9 or 10	/ 20
Total	/ 100

Do not open this booklet until you are told to do so.

Section A: Structured Questions [80 marks]

Answer all the questions in the space provided.

Question 1 [9 marks]

Enzyme X digests a sugar. Figure 1.1 shows the rate of reaction of enzyme X and a mixture of enzyme X and substance Y on the sugar. Both reactions were carried out in a conical flask.

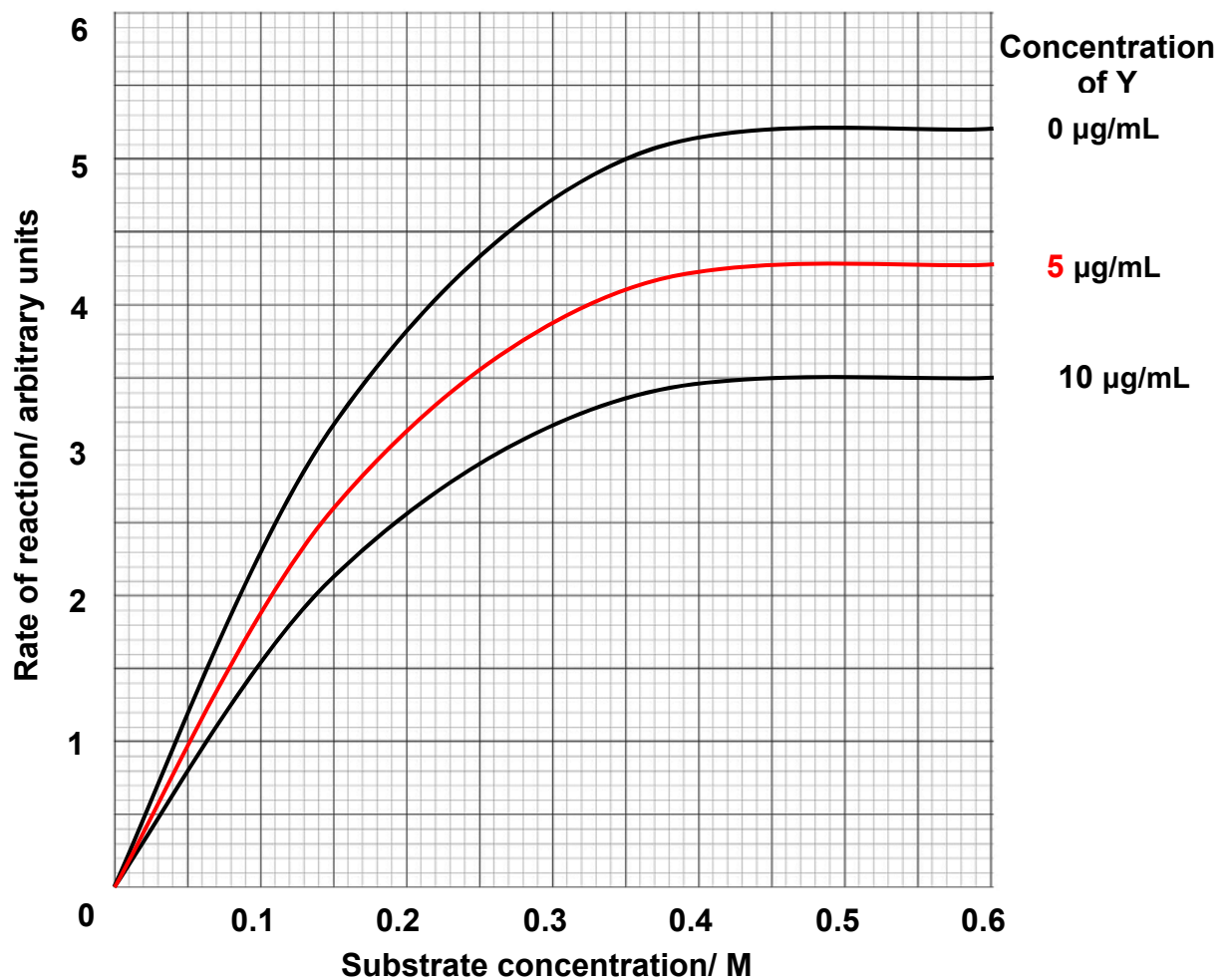


Figure 1.1

(a) (i) Describe and explain the effect of substance Y on the rate of reaction catalyzed by enzyme X. [3]

- a. Substance Y act as a non-competitive inhibitor;
- b. When concentration of Y increases from 0 $\mu\text{g/mL}$ to 10 $\mu\text{g/mL}$, the maximum rate of reaction decreased from 5.2 a.u. to 3.5 a.u.;
- c. Increasing substrate concentration for the reaction with 10 $\mu\text{g/mL}$ of Y does not increase the maximum rate of reaction;
- d. Y has no structural similarity to the substrate;
- e. and binds to allosteric site away from the active site;
- f. This causes a conformational change on the enzyme which alters the shape of the active site;
- g. Hence the enzyme becomes non-functional/no longer able to bind to substrate, and hence decreasing the rate of reaction;

(ii) On Figure 1.1, draw the rate of reaction catalyzed by enzyme X in the presence of **5 $\mu\text{g/mL}$ of Y**. [1]

see graph above.

- Line must be in between the two graphs but no need to be in the middle;
- Label graph;

Enzyme X may be immobilized in alginate beads. Figure 1.2 shows a comparison between the activities of free enzyme X and immobilized enzyme X over a range of temperatures. Equal concentration of free enzyme X and immobilized enzyme X were used.

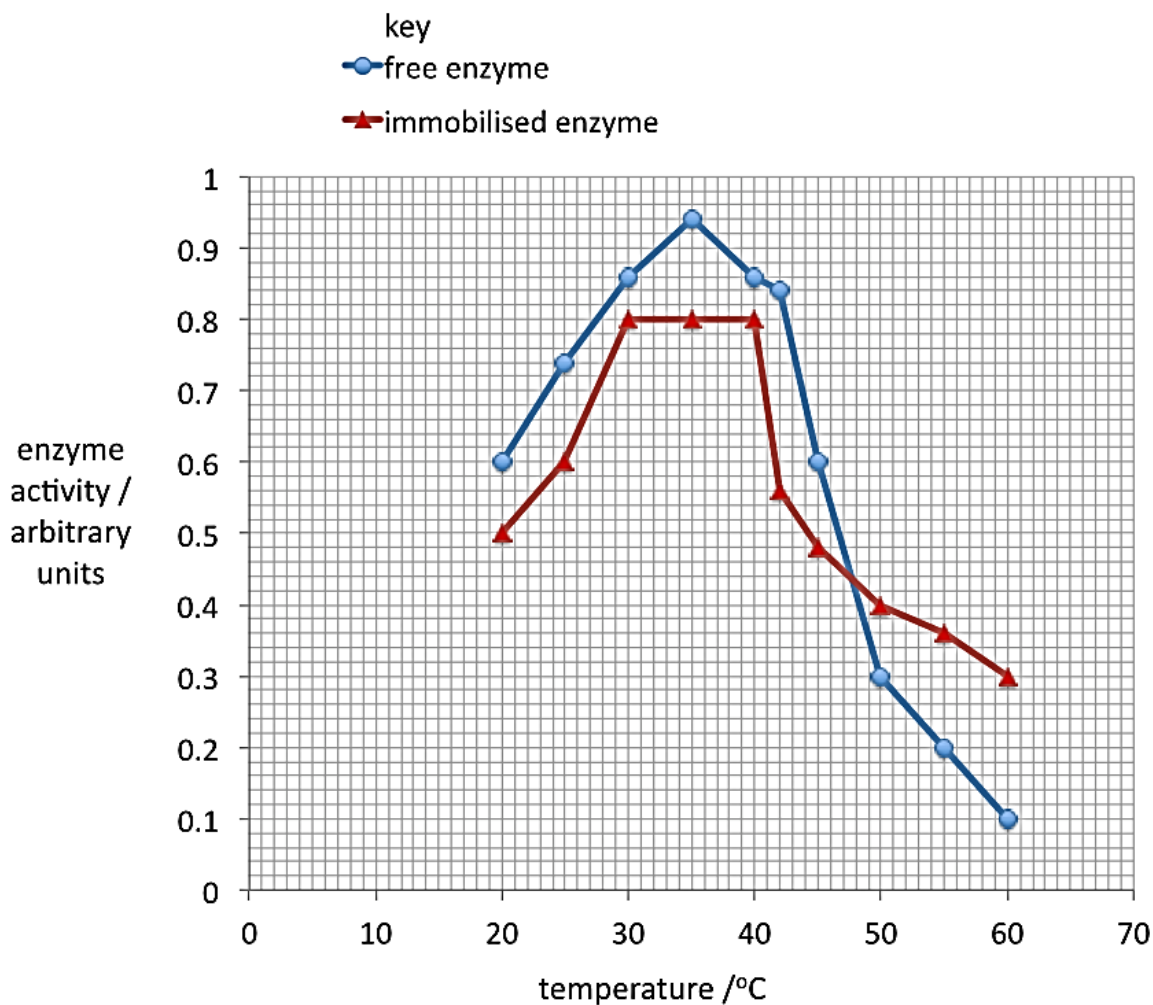


Figure 1.2

(b) With reference to Figure 1.2,

(i) describe and explain the effect of temperature on the activity of free enzyme X.
[3]

- As temperature increases from 20 to 35 °C, the rate of reaction of free enzyme X increases from 0.60 to 0.94 a.u.
- due to the increased kinetic energy of the sugar substrate and enzyme molecules;
- These molecules move faster colliding with one another in correct orientation to form more enzyme-substrate complexes per unit time;
- The increased rate of reaction increases until it reaches its maximum rate of 0.94 a.u. at the optimum temperature of 35 °C when
- most enzyme-substrate complexes formed/ most products formed per unit time;
- Beyond the optimum temperature of 35 °C to 60 °C, rate of reaction decreases from 0.94 to 0.1 a.u.
- as intramolecular bonds are broken and three-dimensional conformation of active sites in enzymes are altered such that substrate can no longer fit

(ii) describe and explain the differences between the activity of free enzyme X and immobilized enzyme X up to 40°C. [2]

- a. From 20 to 40 °C, activity of immobilised enzyme is lower than free enzymes ;
- b. and the maximum activity of immobilised enzyme (0.80 A.U at 30 °C) is lower than that of free enzyme (0.95 A.U. at 35°C/ between 30 to 40 °C);
- c. Enzymes immobilised in alginate beads diffuse out from beads less easily to come into contact with sugar substrate;
- d. less able to move and collide successfully with substrates leading to fewer enzyme-substrate complexes formed;

Question 2 [10 marks]

The control of gene expression functions in a wide range of developmental processes including sex determination.

In *Drosophila melanogaster*, the doublesex (dsx) protein is crucial in the development of sexual phenotype. In males, the presence of dsx-m increases the transcription of genes whose products control male sexual development. The transcription of these genes are repressed in the presence of an isomer dsx-f which is found in females.

Figure 2.1 shows the structure of the dsx-m protein which has a DNA-binding domain and protein-binding domain.

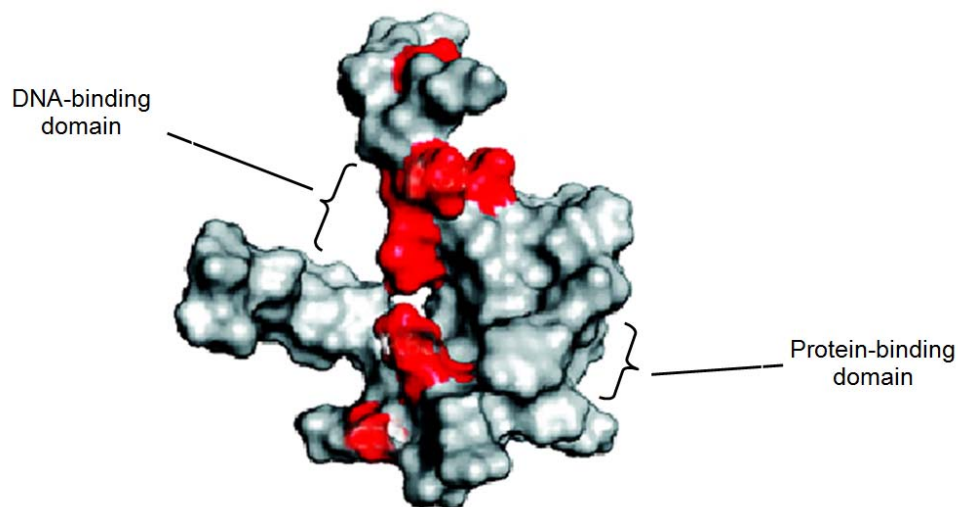


Figure 2.1

(a) State the role of proteins dsx-m and dsx-f. [1]

dsx-m: **activator**;

dsx-f: **repressor**;

Reject: STF

It was found that male *Drosophila* flies showing female characteristics mostly have an amino acid substitution at the DNA-binding domain of the dsx-m protein.

The shaded areas in Figure 2.1 show the positions of amino acid substitutions presumed to affect the DNA-binding domain.

(b) Explain why dsx-m protein contains both a DNA-binding domain and a protein-binding domain. [2]

- a. DNA-binding domain binds to specific **DNA sequences/ nucleotide sequences that make up enhancer**;
- b. Protein-binding domain binds to **other transcription factors / RNA polymerase / DNA bending protein / Mediator protein**;
- c. to stabilize the transcription initiation complex and thus enable it to bind securely on the promoter →;
- d. increase rate of transcription;

(c) Explain why an amino acid substitution within the DNA-binding domain can affect the function of dsx-m protein. [2]

- a. The amino acid substituted has a **different R group with different properties /size/ charge**;
- b. affects **R group interactions / bonds formed** between the amino acids that form the DNA-binding domain;
- c. **tertiary structure/ 3D conformation** of DNA-binding domain no longer;
- d. **complementary to shape of specific DNA sequences/ nucleotides** that form enhancer;

Proteins dsx-f and dsx-m are involved in the regulation of *Drosophila* *yolk protein* gene. The *Drosophila* *yolk protein* gene is expressed in the fat body of adult female under the control of dsx. The presence of dsx-f activates the transcription of *Drosophila* *yolk protein* gene, which allows *Drosophila* *yolk protein* gene to be expressed in the fat body of adult female. The presence of its isomer, dsx-m in males represses the transcription of *Drosophila* *yolk protein* genes.

An *in vitro* transcription under the control of a eukaryotic promoter of *Drosophila* *yolk protein* gene has been developed. The transcription of DNA can occur in:

- **Nuclear extract:** with contents of cell nucleus without genome from a female *Drosophila*
- **Purified system:** with purified RNA polymerase and the following general transcription factors, TFIID (the TATA binding factor), TFIIB and TFIIE (which bind to RNA polymerase).

A series of experiments that compare the efficiency of transcription in the purified system with the efficiency of transcription in a crude nuclear extract have been performed. The two systems were tested with the undeleted template DNA and with various deleted templates that were generated as shown in Figure 2.2. The number represents the number of bases away from the transcription start site (+1).

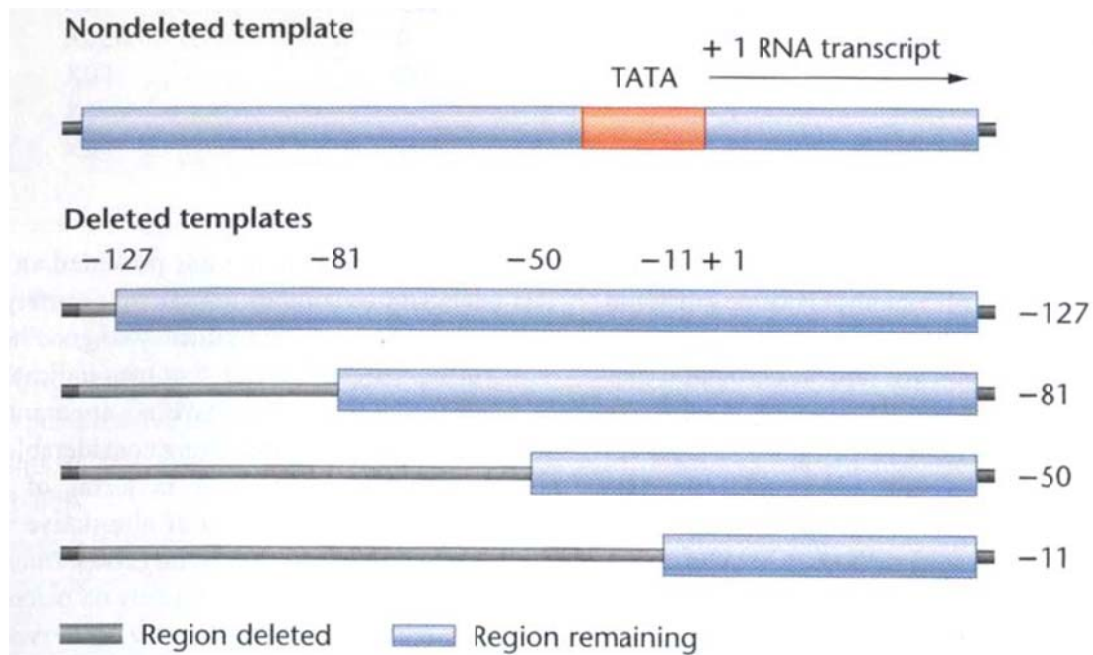


Figure 2.2

Results of the test are shown in Table 2.3.

DNA template added	Nuclear extract	Purified system
undeleted	++++	+
-127 deletion	++++	+
-81 deletion	++++	+
-50 deletion	+	+
-11 deletion	0	0

Table 2.3

Legend:

- + Low efficiency transcription
- ++++ High efficiency transcription
- 0 No transcription

(d) Explain why there is no transcription from the -11 deletion template. [1]

- a. Missing promoter sequence/ missing TATA box;
- b. RNA polymerase cannot bind and transcription initiation complex cannot assemble;

(e) With reference to Table 2.3,

(i) explain the differences in the results between the nuclear extract and the purified system for the undeleted template. [2]

- a. Purified system has only a basal transcription rate;
- b. Nuclear extract has a higher transcription rate;
- c. Due to the presence of specific transcription factors, dsx-F (activator), in nuclear extract
- d. Therefore RNA polymerase has a higher affinity to the promoter region

(ii) explain the difference in the results between the -50 deleted template and -81 deleted template in the nuclear extract. [2]

- a. -50 deleted template has basal transcription rate;
- b. Contained only promoter sequence for general transcription factors (TFIID and TFIIB) and RNA polymerase to bind to;
- c. The binding sites (enhancer) for the rest of the specific transcription factors (Dsx-F activator);
- d. beyond -50 has been deleted

Question 3 [11 marks]

Galactose (*gal*) operon is shown in Figure 3.1. The *gal* operon is regulated negatively by a repressor in a manner comparable to *lac* regulation. The *galactose* operon of *E. coli* consists 3 structural genes - *galE* (epimerase), *galT* (galactose transferase) and *galK* (galactokinase), which are transcribed from two overlapping promoters P_1 and P_2 upstream from *galE*. Regulation of the operon is complex since the *galE* product, an epimerase that converts UDP-glucose into UDP-galactose, is required for the formation of UDP-galactose for cell wall biosynthesis.

When galactose is present, there is a higher rate of transcription to produce enzymes to metabolize galactose. The *gal* operon is controlled by cAMP-CAP as for the *lac* operon. cAMP-CAP binds to region promoter P_1 promoting transcription from P_1 but inhibiting transcription from P_2 . When cells are grown in glucose, basal level transcription occurs from P_2 .

GalR codes for a repressor protein, promoter and operators (O_E and O_I) are present in the *gal* operon.

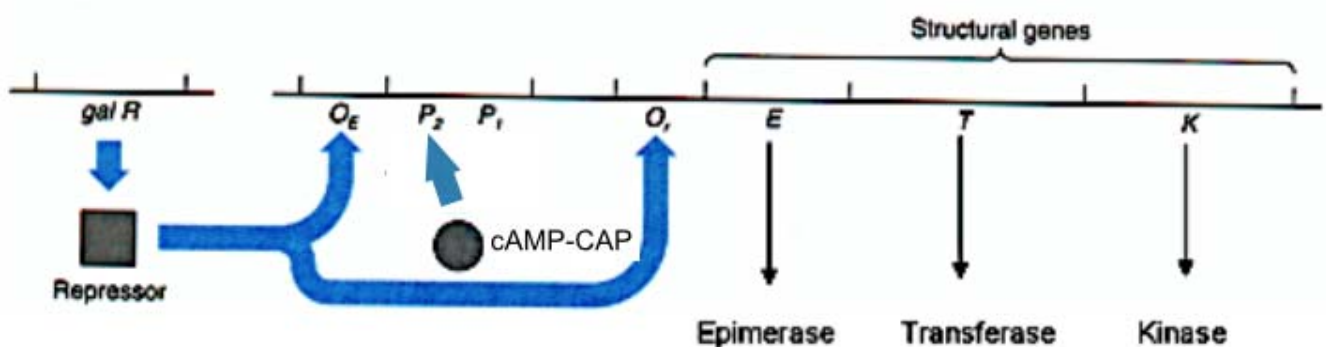


Figure 3.1

(a) In the context of *gal* operon, explain what is meant by the term *operon*. [2]

- a. (Three) structural genes *galE*, *galT* and *galK*; encoding epimerase, galactose transferase and galactokinase respectively;
- b. required for galactose metabolism;
- c. Contain a terminator sequence ;
- d. As a transcriptional unit which is co-ordinately controlled ; (OWTTE)
- e. By a single promoter and operator ;

(b) Using *gal* operon as an example, explain what is meant by *negative control*. [1]

- a. A regulatory mechanism in which the DNA binding regulatory protein is a repressor protein coded by *galR* gene;
- b. that turns off transcription of the *galE*, *galT* and *galK*/structural gene **OR** switch off the operon;

- (c) Explain the function of galactose in this system. [2]
- a. Act as inducer;
 - b. Bind to allosteric site of active repressor;
 - c. Change its conformation;
 - d. and renders it inactive, thus will not bind to the operator region;
- (d) How would expression of the structural genes in *gal* operon be affected if gene for *galR* was missing? [1]
- a. no repressor produced;
 - b. structural genes (rej: operon) will be constitutively expressed;
- (e) Describe how the *gal* operon works under high galactose and low glucose concentration conditions. [4]
- a. (Low concentration of glucose → cyclic AMP accumulates in the cell/ high cAMP.
 - b. Cyclic AMP binds to cAMP receptor protein (CAP);
 - c. CAP complex binds to CAP-binding site within the P1 region;
 - d. increases affinity of the promoter region for RNA pol;
 - e. Increased rate of transcription;
 - f. Presence of inducer (galactose) binds to allosteric site of active repressor ;
 - g. Change its conformation; and renders it inactive thus will not bind to the operator region;
 - h. Thus RNA pol can access to the promoter region;
 - i. Turning on transcription of the structural genes;
- (f) With reference to Figure 3.1, suggest how binding of *gal* repressor prevents the transcription of the structural genes. [1]
- a. Binding of a repressor to the 2 operator causes folding of DNA;
 - b. to prevent access of RNA pol to promoters 1 and 2;

Question 4 [10 marks]

The World Health Organization estimates that 5 million children die each year as a consequence of acute diarrhea. *Escherichia coli* is the cause of a third of cases of childhood diarrhea in developing countries and is also the most prominent cause of diarrhea in travellers to developing countries. *Shigella* species, the cause of bacillary dysentery, taxonomically constitute a subspecies of *E. coli*. Dysentery is an intestinal inflammation, especially in the colon, that can lead to severe diarrhea with mucus or blood in the feces.

Effective treatment and prevention measures are lacking for *E. coli* diarrhea. The mainstay of treatment is oral rehydration. This simple and inexpensive measure has saved countless lives, but it does not influence the natural course of disease and has no intrinsic antibacterial activity. Antibiotic use is of doubtful value since resistance is widespread in *E. coli*, and vaccines are still in the early development phase. Water and sanitation programmes could improve the quality of drinking water, but are prohibitively expensive for many developing countries.

Bacteriophages have been proposed as natural antimicrobial agents to fight bacterial infections in humans, in animals or in crops of agricultural importance. In fact, phages have been used as therapeutic agents in Eastern Europe and the former Soviet Union (and before the antibiotic era in the United States). They have been administered to humans (i) orally, in tablet or liquid formulations, (ii) rectally, (iii) locally (skin, eye, ear, nasal mucosa, etc.), in rinses, and creams, (iv) as aerosols, and (v) intravenously and there have been virtually no reports of serious adverse complications associated with their use.

(a) Identify and explain which bacteriophage would be the preferred therapeutic agent to use in the immediate treatment of dysentery for humans. [3]

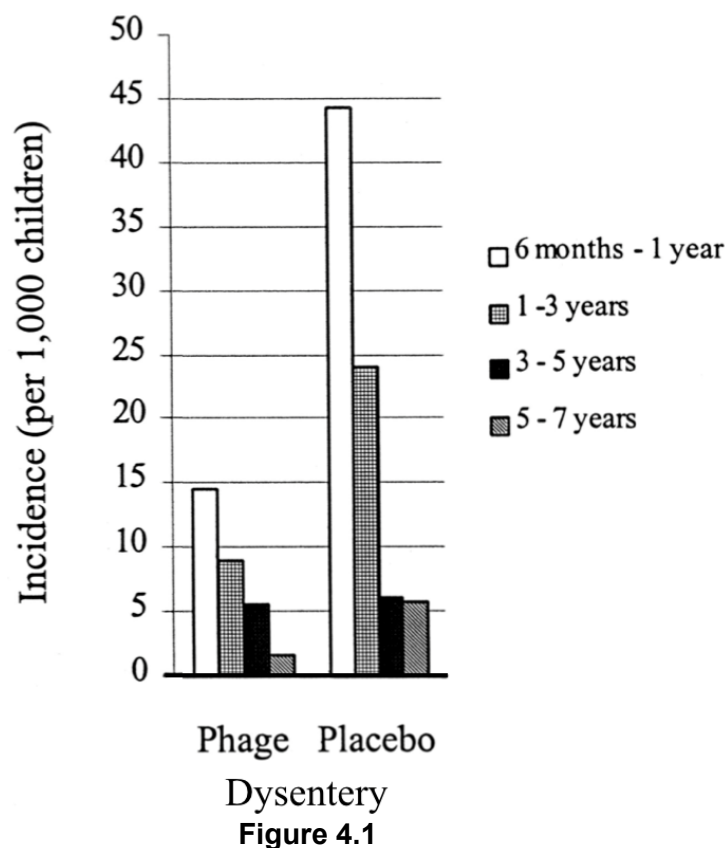
- a. T4 bacteriophage
- b. Adsorption: T4 has tail fibres that act as anti-receptors to allow them bind to receptors on the surface on *E. coli* hosts
- c. T4 is a lytic phage/undergoes lytic life cycle
- d. Synthesis: Late viral protein, T4 lysozyme, coded by phage DNA to break down peptidoglycan of the host cell walls
- e. Release: Kills bacterial host cells via lysis of host cells to release new phage particles
- f. Reduces number of bacteria in the body → Bacteria hosts die → no more diarrhea

(b) Other than the rapid development of resistant *E.coli* variants, suggest why antibiotics may not be the most effective treatment for dysentery. [1]

- a. **Antibiotics kill bacteria rather unspecifically** eg. not just kill *Shigella* sp and can therefore lead to numerous side effects like killing the beneficial *E. coli* that aids digestion in our gut;;
- b. In contrast, species specificity is the rule for phages used in therapy and is commonly quoted as one of the major assets of T4 phage therapy;;
- c. How bacteriophage attaches to correct host cell: Tail fibre act as anti-receptors to recognize specific receptors on host cell surface & bind via weak interactions. Base plate pins strongly anchor phage to host;;

One of the most extensive studies evaluating the utility of therapeutic phages for prophylaxis of infectious diseases was conducted in 1963 and involved phages against bacterial dysentery. A total of 30 769 children (6 months to 7 years old) were enrolled in an oral phage prophylaxis trial against bacterial dysentery. About half of the children living on one side of the streets received, once every week, *Shigella* phages orally in a pressed tablet form, while children on the other side of the streets received a pressed tablet made from sugar alone to act as a placebo. The children were followed for 109 days.

Figure 4.1 shows the incidence of clinical dysentery (diagnosed via appearance of initial symptoms) caused by *E. coli* other than *Shigella* sp. in phage-treated and phage-untreated (placebo) children 6 months to 7 years of age.



(c) (i) With reference to Figure 4.1, comment on the effectiveness of phage therapy.
[3]

- a. Phage therapy is effective;
- b. especially for age groups of 6mths-1yr, 1-3yrs and 5-7 years;
- c. Most effective in 6mths-1yr age group, whereby incidence of dysentery decreased from 44 per 1000 children to nearly 15 per 1000 children when phage treatment was administered;
- d. followed by 1-3 yrs, whereby incidence of dysentery decreased from 24 to 9 per 1000 children;
- e. for 5-7 years, incidence of dysentery decreased from 6 to 1 per 1000 children;
- f. However, no significant difference for age group 3-5yrs as no significant difference between placebo and phage treated subjects;

(ii) Suggest the purpose of the placebo group. [1]

- a. to act as control
- b. to confirm if the reduction in dysentery is due mainly to the effects of phages and not due to other environmental factors

Unlike bacteriophage, the influenza virus is an animal virus. A drug that can inhibit the reproductive cycle of influenza virus is oseltamivir, better known under its trademark name as Tamiflu. Its mechanism of action is illustrated in Figure 4.2 and Figure 4.3.

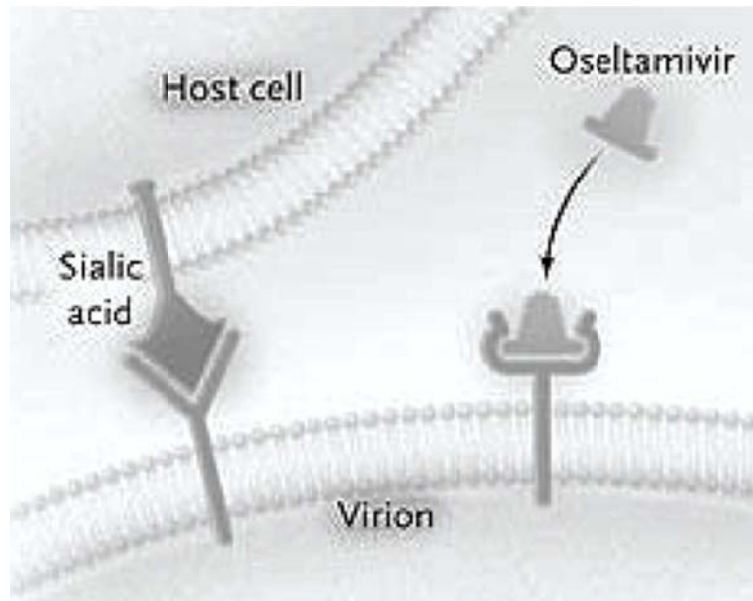


Figure 4.2

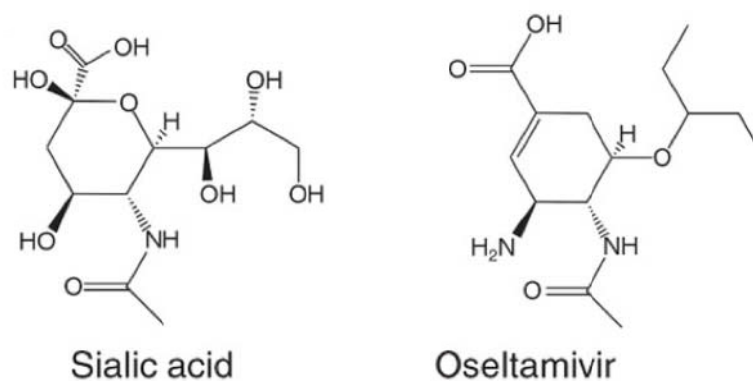


Figure 4.3

(d) Explain how oseltamivir affects the reproduction cycle of influenza virus. [2]

- a. Oseltamivir has similar conformation/spatial and chemical compatibility as sialic acid;
- b. It is thus able to bind to the active site of neuraminidase;
- c. This prevents neuraminidase from cleaving sialic acid
- d. Hence virion will not be able to be released via budding to infect other cells,

Question 5 [10 marks]

The bulbs of onion plants of a particular species display a variety of skin colours. Genetically identical onion plants that produced red onion bulbs were self-pollinated. The colours and number of all the onion bulbs in the progeny were recorded.

Red bulbs 1855
Pink bulbs 634
White bulbs 782

Use the following symbols to represent the alleles:

R red r colourless
P pink p colourless

(a) (i) Draw a genetic diagram in the space below to show the cross described. [4]

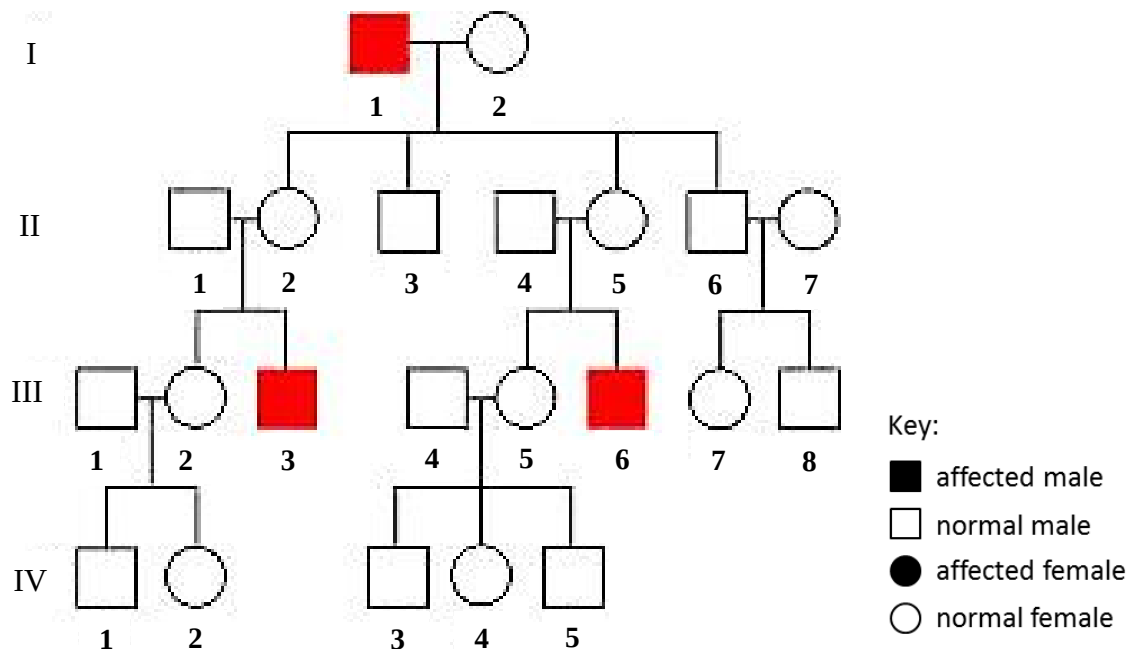
Parental Phenotype	Red bulb		Red bulb		
Parental genotype	PpRr		PpRr		
Meiosis					
Gametes	PR Pr pR pr		PR Pr pR pr		
Offspring genotype and phenotype		PR	Pr	pR	pr
	PR	PPRR red	PPRr red	PpRR red	PpRr Red
	Pr	PPRr red	PPrr pink	PpRr red	Pppr pink
	pR	PpRR red	PpRr Red	ppRR white	ppRr white
	pr	PrRr red	Pppr pink	ppRr white	pprr white
Offspring phenotypic ratio	9 red bulbs: 3 pink bulbs: 4 white bulbs				

- correct genotype;
- correct gametes;
- gamete circle and arrows;
- punnett square correct genotypes;;
- punnett square correct phenotypes;;
- phenotypic ratio;

(ii) Explain the term *epistasis* in this context. [3]

- Epistasis refer to the interaction between two genes that affect skin colour of onion bulb
- whereby a gene at one locus suppresses/masks the phenotypic expression of a gene at a second locus
- Both gene products work in the same pathway resulting in pigmentation of onion.
- Homozygous recessive genotype at both loci or at gene P locus will result in the same phenotype, white bulb.
- Hence homozygous recessive pp is epistatic to R/r alleles.
- The presence of a dominant allele in both loci will complement each other
- Gene product from each locus is required to produce a different phenotype, red bulb.

The pedigree below shows the inheritance of glucose-6-phosphate dehydrogenase deficiency over four generations in an extended family.



(b) Explain the mode of inheritance of this disease that best explains the pedigree shown above. [3]

a. sex-linked recessive disease;

evidence 1 for recessive

- I1 is affected but none of the offsprings are affected (any evidence to show that it is recessive)
- Daughters of I1 and I2 will all inherit the X-chromosome, but none of them show the disease
- thus, diseased allele is recessive as it is masked in the presence of normal allele

evidence 2 for sex-linked

- only males are affected
- Each male only has one X- chromosome inherited from the mother
- Hence, males have a higher probability of being diseased

h. As seen in III3 and III6 where their mothers are a carriers

max 3

Question 6 [10 marks]

In a study on the role of proton transfer through ATP synthase complex, a plant was subjected to flashes of light (indicated by the arrows) and the concentration of ATP synthesized was subsequently recorded as shown in Figure 6.1. The experiment was also conducted in the presence of venturicidin, a potent inhibitor of proton flow in ATP synthase complex.

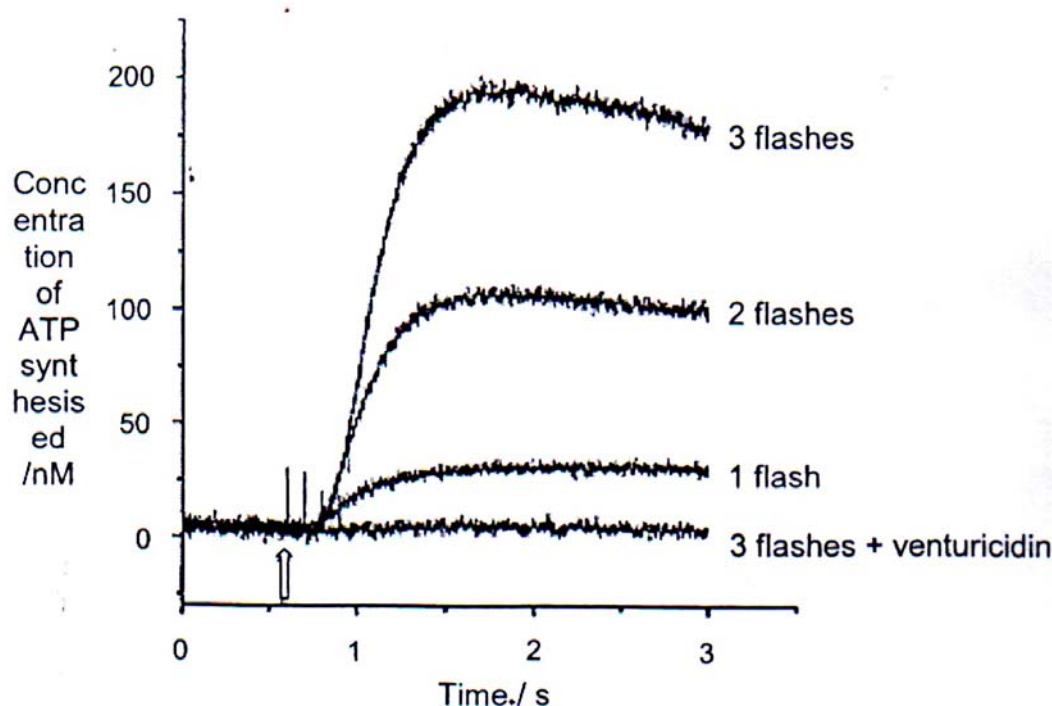


Figure 6.1

- (a) With reference to Figure 6.1,
- (i) Explain the effect of flashing light on ATP synthesis in the absence of venturicidin. [3]
- a. As the number of light flashes increases from 1-3 flashes, the concentration of ATP increases from 25nM-190nM,
 - b. Increasing the number of flashes of light result in an increase in the number of photoexcited electrons from P680 and P700 in PSII & PSI, respectively.
 - c. And results in more excited electrons travelling down the electron transport chain
 - d. Energy release is coupled to pump/actively transport H^+ from the stroma into the thylakoid space/thylakoid lumen
 - e. Resulting in the formation of a proton motive force / electrochemical gradient
 - f. H^+ then diffuses down the electrochemical gradient from the thylakoid space to the stroma via the ATP synthase complex
 - g. Releasing more energy which coupled for synthesise more ATP

(ii) Account for the effect of venturicidin on photosynthesis. [2]

- a. In the presence of venturicidin, ATP concentration remains close to 0nM even when 3 flashes of lights are used
- b. Since ATP synthase is inhibited, H^+ cannot diffuse down the electrochemical gradient
- c. From thylakoid space to stroma
- d. Thus there is no supply of energy to drive ATP synthesis/ no energy released for formation of ATP
- e. This prevents the progression of the Calvin's cycle/ no G3P/no hexose sugar being produced (OWTTE)
- f. Since ATP is not available for conversion of GP to 1,3-bisphosphoglycerate/regeneration of RuBP

A student investigated the pathway by which carbon dioxide is converted to organic compounds during photosynthesis. He used the apparatus shown in Figure 6.2.

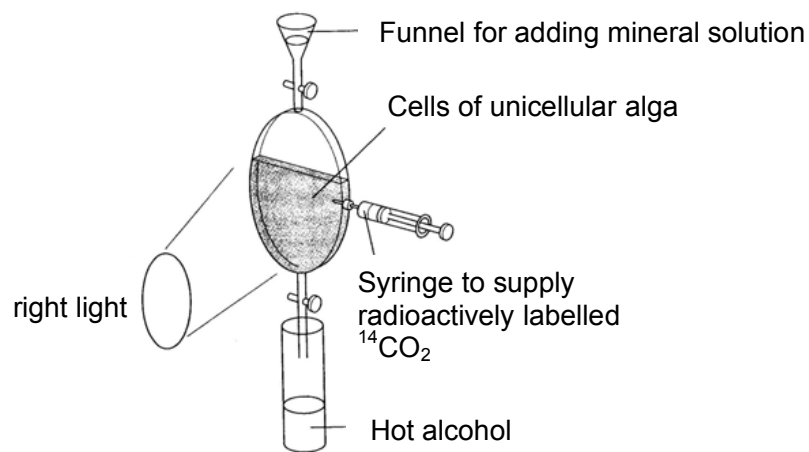


Figure 6.2

While the apparatus was in a dark room, he supplied the algal cells with $^{14}CO_2$. The contents of the apparatus were thoroughly mixed and the light was switched on subsequently. At five-second intervals, he released a few of the cells into hot alcohol, which killed the cells very quickly. The results were shown in the following chromatograph (Figure 6.3).

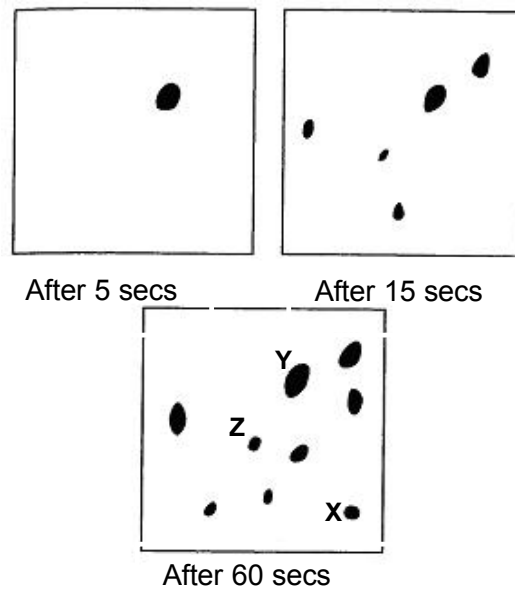


Figure 6.3

(d) (i) One round of Calvin cycle takes about 60s to complete. The intermediates of the Calvin cycle were subsequently analysed to be ribulose-1,5-bisphosphate, glyceraldehyde-3-phosphate and 3-phosphoglycerate. With reference to Figure 6.3, identify X, Y and Z and explain your answer. [4]

X – ribulose-1,5-bisphosphate;
 Y – 3-phosphoglycerate;
 Z – glyceraldehyde-3-phosphate;

- Sequence of appearance of radioactive spots is Y (3-phosphoglycerate) which appeared after 5s, Z (glyceraldehyde-3-phosphate) appearing 10s later and X (ribulose-1,5-bisphosphate) appearing after 60s.
- Phase 1: (Carbon Fixation) - Each radioactive CO_2 molecule is incorporated to 5C ribulose bisphosphate (RuBP) ;
- The product of the reaction is a six-carbon intermediate splits into two molecules of radioactive 3-phosphoglycerate for each CO_2 fixed.
- Phase 2: (Reduction) - Each molecule of 3-phosphoglycerate receives an additional phosphate group from ATP, forming radioactive 1,3-bisphosphoglycerate as a product.
- Next a pair of electrons donated from NADPH reduces 1,3-bisphosphoglycerate to glyceraldehyde-3-phosphate (G3P/TP) = 3C sugar
- Phase 3: (Regeneration) The other 5 molecules must be recycled to regenerate the 3 molecules of radioactive RuBP using ATP.

- (iii) Suggest how increasing carbon dioxide concentration will affect the time of appearance of X, Y and Z on the chromatograph, [1]
- a. CO_2 is a limiting factor of photosynthesis (Calvin Cycle)
 - b. Increasing CO_2 will increase rate of appearance of X, Y and Z on the chromatogram as rate of CO_2 fixation will increase.

Question 7 [10 marks]

Ras is a GTP-binding protein that helps to broadcast signals from the cell surface to other parts of the cell to stimulate cell proliferation. If Ras function is inhibited, the cell proliferation responses normally induced by the activated receptors do not occur. Conversely, if a hyperactive mutant, Ras protein, is introduced into some cell lines, the effect on cell proliferation is the same as that induced by the binding of ligands to cell surface receptors. This hyperactive product of mutant Ras gene promotes the development of cancer. Figure 7.1 shows the sequence of events in a Ras cell cycle-stimulating pathway.

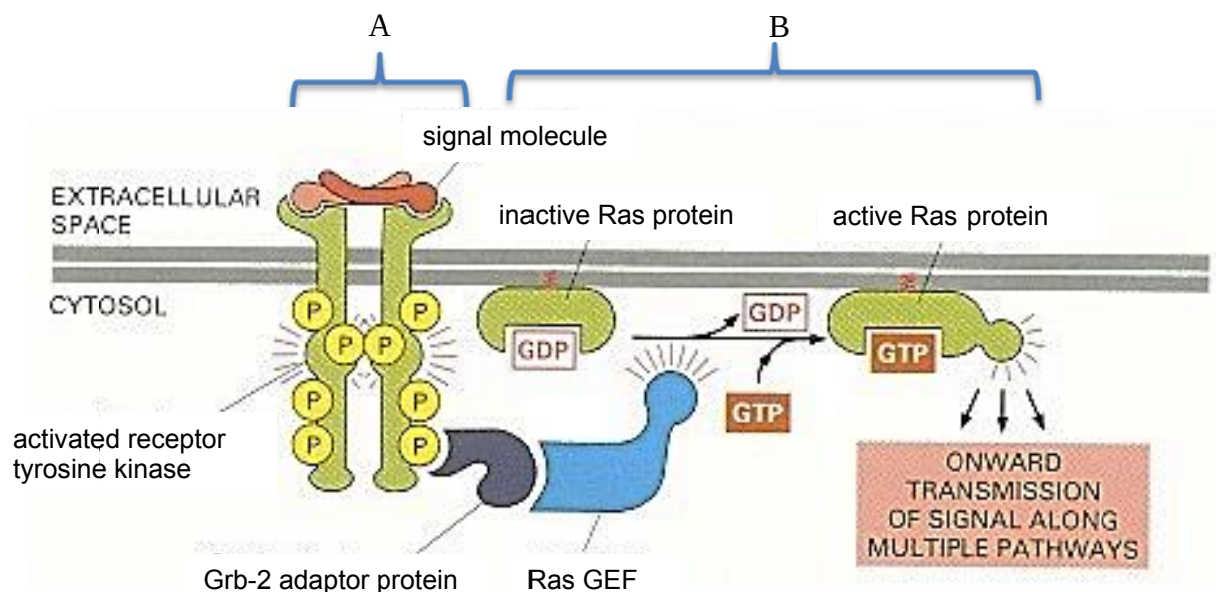


Figure 7.1

(a) (i) Briefly describe the stages A and B shown in Figure 7.1. [4]

Stage A

- binding of signal molecule to the 2 receptors causes 2 receptor subunits / polypeptide chains / monomers / molecules to form a dimer / dimerise;
- cross phosphorylation of tyrosine amino acids on each other's tyrosine amino acid tails
- activation of the receptor tyrosine kinase region leads to
- and subsequent auto-phosphorylation residues on their respective tail of tyrosine amino acid subunits on the rest of the cytoplasmic tails
- activated tyrosine are docking sites with high affinity for docking/relay proteins

Max 2

Stage B

- relay protein(s), Grb-2 adaptor protein recognizes and binds to specific phosphorylated tyrosine(s) which acts as docking protein
- for Ras GEF which becomes activated
- which displaces GDP with GTP in ras protein
- which activates G protein

- j. other proteins or enzymes to bring about various cellular responses (= cell proliferation).

Max 2

(ii) Ras protein is a plasma membrane bound G-protein. Suggest how the activated G-protein can relay the signal from the ligand to activate cellular response. [2]

- a. By activating enzymes such as adenylyl cyclase which will catalyse
- b. production of second messengers, such as cAMP
- c. cAMP will activate protein kinase A
- d. resulting in a phosphorylation → activate proteins necessary for cellular responses

The receptor for epidermal growth factor (EGF) can be activated by a deletion that removes part of its extracellular domain. These mutant receptors are able to form active dimers even in the absence of EGF, and thus they produce a stimulatory signal inappropriately. Besides mutation to the gene that codes for the extracellular domains of the receptors, 30% of human tumors have a hyperactive Ras mutation.

(iii) With reference to Figure 7.1, suggest how mutation in the Ras gene may bring about cancer. [3]

- a. Ras gene is a proto-oncogene which codes for the Ras protein.
- b. Most are gain of function/dominant mutations → hyperactive ras mutations where mutant forms of Ras become oncogene
- c. Point mutation to the domain that codes for GTPase (GTP cannot be hydrolysed to GDP)
- d. eg. Base pair substitution
- e. no longer codes for functional GTPase and is permanently bound to the GTP
- f. Can no longer be turned off even when ligand/growth factor is no longer bound to the receptor tyrosine complexes
- g. The last kinase activates a transcription factor that turns on genes for one or more proteins that stimulate the cell cycle. Activate other cellular proteins which stimulate cell division continuously and are no longer density and anchorage dependent → tumor

(iv) Besides the abovementioned mutation, identify and suggest how a virus may cause Ras proto-oncogene to become an oncogene. [1]

- a. The provirus (the integrated DNA copy of the retroviral RNA) may integrate by chance near one of the cell's normal proto-oncogenes;
- b. The strong promoters and enhancers in the provirus then stimulate high levels or inappropriate timing of transcription of the proto-oncogene, leading to stimulation of host cell proliferation;

OR

- c. A retrovirus may pick up a copy of the host proto-oncogene and integrate it into its genome when it infects the first host.
- d. The new viral oncogene may be mutated during the process of transfer into the virus, or may be expressed at abnormal levels because it is now under the control of viral promoters.
- e. Retroviruses that carry these oncogenes can infect and transform new normal host cells into tumour cell.

Question 8 [10 marks]

The Hawaiian islands possess one of the most remarkable insect faunas on Earth and are considered as natural laboratories for the study of the process of speciation and evolution.

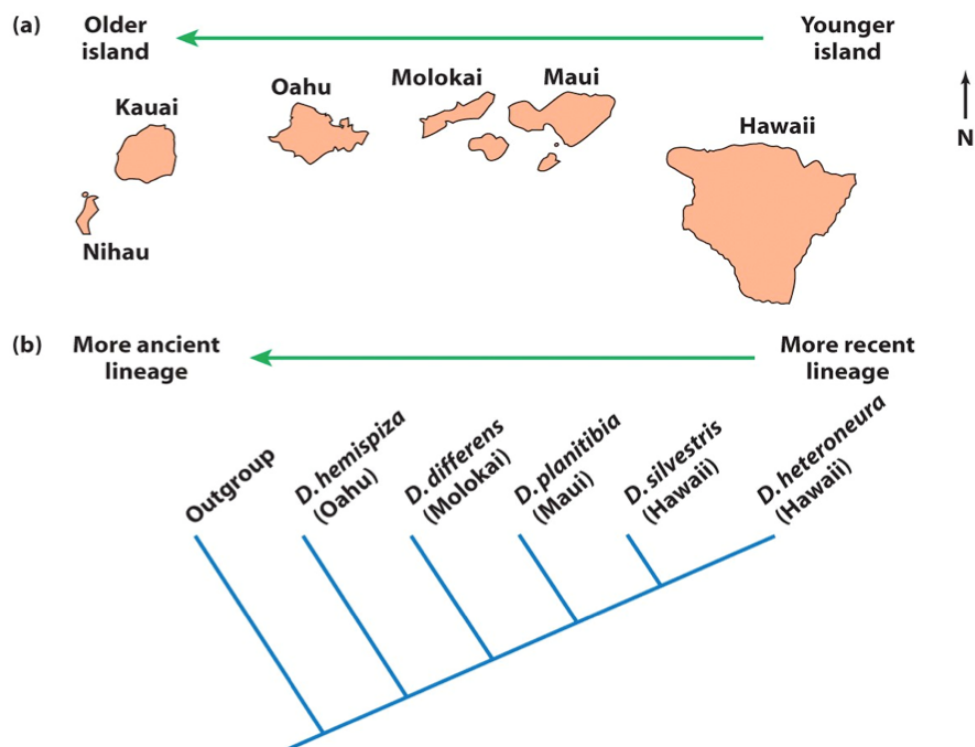


Figure 8.1 (a) Illustration of distribution and geological age of Hawaiian islands. (b). Distribution of some *Drosophila* species across the islands in correspondence how recent the species had been formed.

An example of evolution of a large number of divergent forms from a primitive ancestor in these islands can be observed from high species diversity in the genus *Drosophila* (Figure 8.1(b)). These *Drosophila* are also collectively known as Hawaiian *Drosophila*. Across the islands hold at least 411 species of *Drosophila*, with characters ranging elaborately from size, pigmented wings to different mouthparts and unique ornaments on the head (Figure 8.2). While some of these characters appear to be advantageous for survival in species-specific niches, other characters are thought to play a role in courtship behavior.

Throughout their evolutionary history, Hawaiian *Drosophila* have repeatedly colonized islands or subdivided populations within an island.

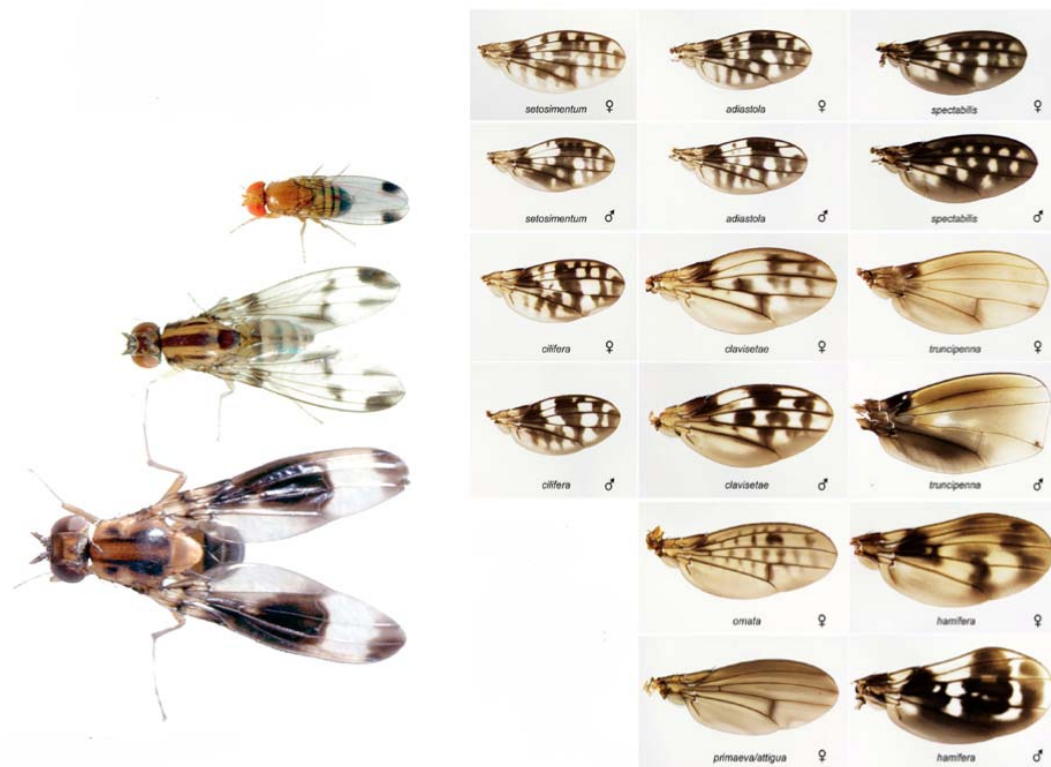


Figure 8.2

- (a) Explain how the evolution of different species in the Hawaiian Islands could have occurred. [6]

Founder effect (with ref to the different islands)

- Founder effect happens when a small group of *Drosophila* from a population colonises another island or a different niche
- Changes in allele frequencies are significant from one generation to the next due to genetic drift

Natural selection

- Variation exists in the population and different selection pressures on different islands lead to different traits being selected for/different traits at a selective advantage
- For e.g, different mouthparts are favoured based on type of food available in niche
- Leading to differential survival and reproductive rates among different individuals
- Causing a change in allele frequency within the population.
- This also leads to adaptive radiation as seen in the variety of characters across species in *Drosophila* genus.

Allopatric speciation

- Islands causes geographical isolation to occur
- Therefore there is no gene flow as there is no interbreeding between populations on different islands
- Over time, mutations accumulate within each population leading to further changes in allele frequency → formation of new species.
- This is known as allopatric speciation.

Sympatric speciation

- l. Within an island, populations may subdivide and evolve different traits that serve as pre-zygotic barriers
- m. such as ecological isolation and behavioural isolation (courtship rituals).
- n. causing sympatric speciation → different species being formed.

max 6

Differences in the cytochrome b DNA sequence of several species of *Drosophila* from different islands were measured and plotted against time since divergence from the primitive ancestor (MYA) as seen in Figure 8.3.

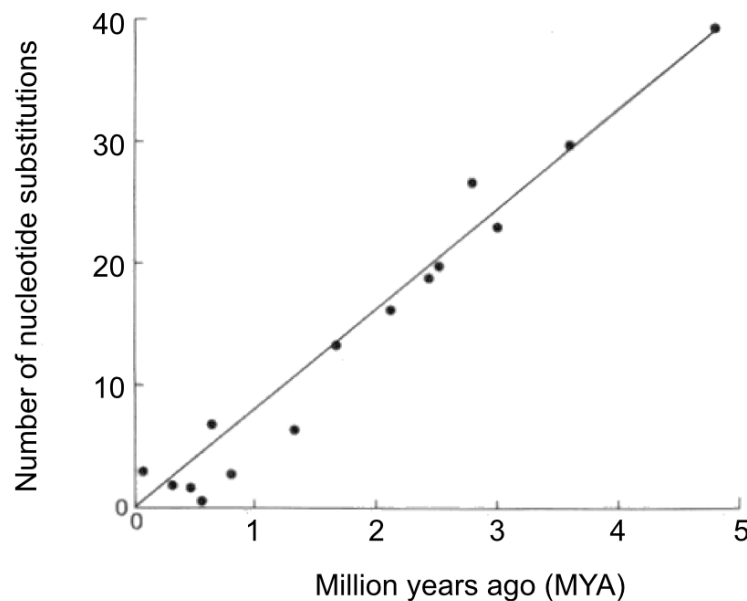


Figure 8.3

(b) (i) Describe how these changes support neutral theory of molecular evidence. [3]

- a. Changes in the nucleotide sequence arises through neutral mutation
- b. for e.g. silent mutation or missense mutation where the change in amino acid does not occur in a critical region of the enzyme.
- c. There is no effect on the phenotype and fitness of organism and thus allowed to accumulate
- d. Small number of changes over million of years,
- e. indicating that rate of mutation is slow as cytochrome b gene is a crucial gene in living organisms
- f. The plot of the line is straight,
- g. indicating that the rate of mutation is constant

(ii) Suggest a challenge in using the molecular clock. [1]

- a. No gene marks time with complete precision. In fact, some portions of the genome appear to have evolved in irregular fits and starts that are not at all clocklike.

- b. Even genes that seem to have reliable molecular clocks are accurate only in the statistical sense of showing a fairly smooth average rate of change. Over time, there may still be chance deviations above and below that average rate.
- c. The same gene may also evolve at different rates in different groups of organisms, making it necessary to calibrate and use molecular clock with care.
- d. Even among genes that are clocklike, the rate of the clock may vary greatly from one gene to another; some genes evolve a million times faster than others.

award 1 mark for any point above

Section B [20 marks]

Answer **only ONE** question.

Write your answers on the separate answer paper provided. Your answers should be illustrated by large, clear labeled diagrams, where appropriate. Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

Question 9

(a) Explain the importance of lipids in living organisms. [7]

(a) Good energy storage;

1. Has 3 long hydrocarbon chains/proportionately more H atoms compared to oxygen
2. Release more hydrogen ions when oxidized during respiration
3. Allows production of twice the amount of energy as compared to same unit mass of carbohydrates
4. Yields twice the amount of metabolic water when oxidized during respiration

(b) Weight-efficient compared to carbohydrates;

1. due to lower density/molecular mass
2. Favours locomotion of animals as they can be stored in large amounts in adipose tissues.
3. Favours seed dispersal since lightweight as it can be stored in large amounts in the seed endosperm.

(c) Provides buoyancy for aquatic animals;

1. blubber & fats are less dense than carbohydrates

(d) Cushions against shock;

1. aggregate together in large quantities in adipose tissues away from the aqueous environment into compact forms.
2. Can surround and protect vital organs on a long term basis to absorb mechanical shock in case of accidents or falls

(e) Good heat insulators;

1. aggregate together in large quantities in adipose tissues away from the aqueous environment into compact forms under the surface of the skin

(f) Good electrical insulators;

1. non-polar and thus repels ions -> high electrical resistance
2. Ensures rapid transmission of nerve impulses in myelinated nerve cells which run long distances along length of body

(b) Compare and contrast between collagen and cellulose. [8]

	Collagen	Cellulose
a. Variety in monomer	Glycine, proline and hydroxyproline amino acids	Only B-glucose molecules
b. Bonds between monomers	Peptide bonds formed between amino acid monomers	B 1,4, glycosidic bonds formed between B-glucose monomers
c. Structure adopted by each chain	Loose helical structure of polypeptide chain due to the lack of intra-chain H bonds (because of lack of N-H group in proline)	Linear chain due to rotation of alternate glucose residue, resulting in bulky side chains on each side of the chain
d. Subunit making up macromolecule	Tropocollagen – 3 polypeptide chains wound tightly together	Linear unbranched chain
e. Arrangement of subunit in macromolecule	Tropocollagen arranged in staggered out of step manner to increase tensile and mechanical strength	Long chains run parallel to each other, but not in a staggered manner
f. Bonds involved in stabilizing macromolecule	Covalent cross-linkages formed between the C-terminus of one tropocollagen and the N-terminus of another tropocollagen	Cumulative strength of H-bonds formed between the hydroxyl groups projected outwards in all directions of each chain
g. Organism	Produced in animals	Produced in plants
Similarities		
h.	Both serve as structural support with high tensile and mechanical strength	
i.	Both are insoluble in water due to the lack of hydroxyl groups to form H-bonds with water.	

(c) Enzymes are prominent examples of globular proteins. Describe the mode of action of an enzyme. [5]

Mode of action:

- Precise active site to which the substrate molecules with complementary conformation and compatible chemical groups become attached.
- Collision between substrate and enzyme at the correct orientation causes the substrate molecules to bind to the enzyme molecule at its active site to form an enzyme-substrate (E-S) complex.
- Enzyme and substrate are held together by weak bonds, such as ionic, hydrogen and hydrophobic bonds, which can be made and broken rapidly.

Induced-fit hypothesis:

- Initially, the active site does not exist in a conformation that is exactly complementary to the substrate.
- As the substrate enters the active site, the amino acids which make up the active site are moulded or 'induced' into a precise shape complementary to the substrate

- f. This brings a greater interaction between the chemical groups of the substrate and active site. (greater chemical and spatial compatibility)
- g. Enables the enzyme to perform its catalytic function more efficiently.
- h. For example, in reactions involving 2 or more reactants, the active site brings the substrates together in the proper orientation for the reaction to occur
- i. As the active site clutches the substrates, the enzyme may stress the substrate molecules, stretching and bending critical chemical bonds within the molecule that must be broken during the reaction.
- j. The active site may also provide a microenvironment that is conducive to a particular type of reaction.

Question 10

(a) Explain how the structure of neuron facilitates impulse transmission. [7]

	Structure	Function
a.	Dendrites	Extensively branched so that many synapses can be made with many other neurons
b.	Numerous mitochondria in cell body and axon	Produce ATP for sodium-potassium pumps / transport of vesicles / synthesis of proteins
c.	Axon	Conducts impulses away from cell body
d.	Nissl's granules	Synthesis of proteins such as $\text{Na}^+\text{-K}^+$ pump / acetylcholinesterase
e.	Myelin sheath	Acts as an electrical insulator, preventing the formation of action potentials
f.	Nodes of Ranvier	Enable action potential to jump from node to node, increasing the speed of transmission of a nerve impulse
g.	Presence of synaptic vesicles in synaptic knob	Synaptic vesicles contain neurotransmitters for synaptic transmission
h.	$\text{Na}^+\text{-K}^+$ pump	Pumps 3 Na^+ out of in exchange for 2 K^+ into axoplasm for maintenance of resting potential
i.	Voltage-gated Na^+ / K^+ / Ca^{2+} channel	Allows for influx of Na^+ / efflux of K^+ / efflux of Ca^{2+} upon trigger by change in membrane voltage
j.	Chemically-gated Na^+ channel	Allows for diffusion of Na^+ upon binding of neurotransmitter
k.	AVP	

- (b) Describe how unidirectional transmission of nerve impulses is ensured along a neuron and across a synapse. [8]

Along a neuron (max 5)

- a. Ref to AP/repolarisation phase
- b. → absolute refractory period
- c. where no stimulus, however strong, can initiate another AP
- d. where the membrane cannot respond to another normal stimulus/be depolarised;
- e. prolonged efflux of K^+ results in hyperpolarisation;
- f. Leading to relative refractory period;
- g. when another AP can only be initiated if the stimulus is stronger than normally required
- h. when Na^+ moves to the region behind the node, membrane will require a greater depolarisation than normal to reach threshold potential → no AP
- i. while Na^+ that moves forward will cause depolarisation and positive feedback and hence AP at region ahead
- j. Ensures that the AP is propagated to the node ahead towards the synaptic terminal;

Across a synapse (max 3)

- k. Synaptic vesicles/neurotransmitters present only in pre-synaptic terminals
- l. Can only release neurotransmitter from pre-synaptic knobs
- m. Numerous mitochondria on presynaptic knobs to allow for exocytosis of neurotransmitters and endocytosis of ethanol and choline to re-form acetylcholine at the knobs;
- n. Neurotransmitter receptors on post-synaptic membrane only
- o. ligand-gated Na^+ ion channels are present only on the post-synaptic membrane / dendrites / cell body of the post-synaptic neuron;
- p. Acetylcholinesterase present in cleft/at post-synaptic membrane to hydrolyse neurotransmitter to maintain diffusion gradient;
- q. Hence impulse/neurotransmitter is transmitted only from pre-synaptic terminal to post-synaptic neuron;

(c) Describe how two named membrane proteins are involved in impulse transmission along the axon of a motor neuron. [5]

(a) Sodium and potassium leak channels;

1. there are more K⁺ leak channels than Na⁺ leak channels,
2. leading to more diffusion of K⁺ than Na⁺ down their respective concentration gradient via facilitated diffusion
3. this results in a net loss of positive charge
4. Resting membrane potential of neurons restored/ maintained at about -70mV, so that it can be excited to produce action potential when there is a stimulus

(b) sodium potassium pumps;

1. Three sodium ions are pumped out of the neurone while two potassium ions are pumped into the neuron
2. via active transport against their respective concentration gradient
3. it restores the original ionic gradients and results in a net loss of positive charges
4. Resting membrane potential of neurons restored / maintained at about -70mV, so that it can be excited to produce action potential when there is a stimulus

(c) Voltage-gated sodium channels;

1. A stimulus causes some voltage-gated Na⁺ channels to open, which allows influx of Na⁺
2. and results in depolarization of membrane
3. If the depolarization reaches threshold potential of -55mV, an action potential is generated, so that more / all voltage-gated Na⁺ channels will open
4. Depolarising the membrane potential to +40mV

(d) Voltage-gated potassium channels;

1. At the peak of an action potential, voltage-gated K⁺ channels opens, which allows efflux of K⁺
2. and hence repolarisation
3. Voltage-gated K⁺ channels are slow to close,
4. so that excessive K⁺ efflux causes the membrane potential to hyperpolarized